A CASE-STUDY BASED REPORT

ON

Credit Card Fraud Detection

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ABSTRACT

Banking systems process a large number of financial transactions every day, making the detection of fraudulent transactions an important challenge. Traditional methods may not always be effective in identifying changing patterns of fraudulent activity. This project, Velora Bank Fraud Intelligence, presents a machine-learning-based bank fraud detection and workflow simulation system designed to identify potentially fraudulent transactions

The system uses a synthetic dataset containing 2,000 transaction **records** with features such as transaction amount, transaction hour, weekend indicator, merchant category, transaction type, country, international transaction status, customer average transaction amount, recent transaction count, and device risk. A Random Forest Classifier is used for fraud prediction after preprocessing the numerical and categorical features. The preprocessing pipeline includes missing-value handling, feature scaling, categorical encoding, and smote to address class imbalance. The trained model generates a fraud probability and prediction for each transaction.

The system further combines machine-learning predictions with transaction and behavioral risk analysis to calculate an overall risk score. Based on the risk level, transactions are classified as low, medium, or high risk. Suspicious transactions can be held for customer verification, while customers can review transactions and report suspected fraud. Bank administrators can investigate reported cases, add investigation notes, resolve fraud cases, manage card status, and maintain investigation records.
The project demonstrates the practical application of machine

learning in fraud detection by integrating predictive modelling, transaction monitoring, behavioral analysis, risk scoring, alerts, and investigation workflows into a single system. The application is developed as an educational and demonstration system and does not process real banking transactions or connect to real financial systems

INTRODUCTION

Background

The rapid growth of digital banking and online financial transactions has increased the need for effective fraud detection systems. Banks process a large number of transactions involving different customers, merchants, transaction types, locations, and devices. Identifying fraudulent transactions manually can be difficult because suspicious activities may appear similar to legitimate transactions. Therefore, Machine Learning can be used to analyse transaction patterns and identify transactions that have a higher probability of being fraudulent.

Velora Bank Fraud Intelligence is a Machine Learning-based bank fraud detection and workflow simulation system. The project focuses on analysing transaction-related information and predicting whether a transaction is potentially fraudulent. The system uses a **Random Forest Classifier** along with preprocessing techniques and **SMOTE** to handle the transaction data and class imbalance.

The project goes beyond simply predicting fraud. The Machine Learning prediction is combined with transaction and behavioural risk analysis to generate an overall risk score. Based on this score, transactions are classified into different risk levels, allowing suspicious transactions to be identified and reviewed through the application.

Motivation for the Project

The main motivation for this project is the need to identify suspicious banking transactions quickly and systematically. Fraudulent transactions can cause financial losses and may affect customer trust in banking services. A system that automatically analyses transaction characteristics and highlights potentially fraudulent activity can help reduce the effort required for manual monitoring.

Another motivation is to demonstrate how Machine Learning can be integrated with a complete fraud-management workflow. Instead of treating fraud detection only as a classification problem, this project connects prediction with risk scoring, transaction monitoring, customer verification, fraud reporting, and administrator investigation.

Overview of the Work Carried Out

The project was developed as an integrated fraud detection and workflow simulation system. The major work carried out includes:

1. **Data Preparation:** Transaction data containing numerical and categorical attributes was prepared for Machine Learning.
2. **Data Preprocessing:** Missing values, numerical features, and categorical features were handled using an appropriate preprocessing pipeline.
3. **Class Imbalance Handling:** SMOTE was applied to address the imbalance between fraudulent and legitimate transaction classes.
4. **Machine Learning Model:** A Random Forest Classifier was trained to predict potentially fraudulent transactions.
5. **Risk Analysis:** Machine Learning predictions were combined with transaction and behavioural factors to calculate an overall risk score.
6. **Fraud Workflow:** The system supports transaction monitoring, suspicious-transaction handling, customer verification, fraud reporting, and administrator investigation.
7. **Web Application:** The functionality is provided through a browser-based application with separate customer and administrative workflows.

This combination of Machine Learning prediction and fraud-management workflows makes the project a practical demonstration of how intelligent fraud detection can be incorporated into a banking environment.

PROBLEM STATEMENT

Case-Study Scenario

Consider a bank customer who regularly performs online transactions using their bank account. Most of the transactions are legitimate, but occasionally a transaction may show unusual characteristics, such as an unexpectedly high amount, an unusual transaction time, an international transaction, or a high-risk device. If these transactions are not identified quickly, they may result in financial loss to the customer.

A conventional system that relies only on fixed rules may not effectively identify different combinations of suspicious transaction characteristics. Therefore, there is a need for a system that can analyse multiple transaction and behavioural features and identify transactions that have a higher probability of being fraudulent.

Velora Bank Fraud Intelligence addresses this problem by using Machine Learning to analyse transaction data, predict potential fraud, calculate an overall risk score, and support further investigation through a simulated banking workflow.

Central Problem Statement

“How can Machine Learning be used to identify potentially fraudulent bank transactions by analysing transaction and behavioural characteristics, while providing a risk-based workflow for verification and investigation?”

Formal Problem Definition

Let each transaction be represented by a set of transaction and behavioural features such as transaction amount, transaction time, merchant category, transaction type, country, international transaction status, customer transaction behaviour, recent transaction count, and device risk.

The objective is to learn a Machine Learning model that takes these transaction features as input and predicts whether the transaction is potentially **fraudulent or legitimate**. The prediction is then used along with additional risk factors to calculate an overall transaction risk level.

The system therefore performs the following major tasks:

1. Preprocess the transaction data.
2. Handle the imbalance between fraudulent and legitimate transactions.
3. Train a Random Forest classification model.
4. Predict the probability of fraud for a transaction.
5. Calculate an overall risk score using fraud probability and relevant risk factors.
6. Classify the transaction into LOW, MEDIUM, or HIGH risk.
7. Support verification, fraud reporting, and administrator investigation for suspicious transactions.

Objectives

- To develop a Machine Learning-based system for detecting potentially fraudulent bank transactions.
- To preprocess numerical and categorical transaction features effectively.
- To use SMOTE for handling class imbalance in the training data.
- To train and use a Random Forest Classifier for fraud prediction.
- To generate transaction-level fraud probabilities and risk levels.
- To incorporate behavioural and transaction-related factors into risk analysis.

- To provide a workflow for customer verification and fraud reporting.
- To provide administrative functionality for investigating and resolving suspected fraud cases.

Assumptions and Limitations

Assumptions

- The available transaction dataset contains sufficient information for training the fraud detection model.
- The transaction and behavioural features used by the system provide useful indicators of potential fraud.
- The Machine Learning model can identify patterns from the available training data.
- The system is used as a simulation for educational and demonstration purposes.

Limitations

- The system uses a synthetic transaction dataset rather than real banking transaction data.
- The model's performance depends on the quality and characteristics of the available dataset.
- Fraud patterns may change over time, requiring the model to be retrained with updated data.
- The system does not connect to real banking systems or process real financial transactions.
- The fraud detection result is a prediction and should not be treated as a guaranteed determination of fraud.

DATASET AND EXPLORATORY DATA ANALYSIS

About the Dataset

The dataset used in the Velora Bank Fraud Intelligence project is a synthetically generated transaction dataset created specifically for training and testing the fraud detection model. The training pipeline generates 2,000 transaction records using predefined transaction and behavioural patterns. The dataset is designed to simulate realistic characteristics that can be associated with potentially fraudulent banking transactions.

Each transaction contains numerical and categorical attributes that are used as input features for the Machine Learning model.

Attribute	Type	Description
amount	Numerical	Monetary value of the transaction
transaction_hour	Numerical	Hour at which the transaction occurred
is_weekend	Numerical/Binary	Indicates whether the transaction occurred on a weekend
merchant_category	Categorical	Category of the merchant, such as food, travel, luxury or general
transaction_type	Categorical	Type of transaction, such as online purchase, card-present or cash withdrawal
country	Categorical	Country associated with the transaction

Attribute	Type	Description
is_international	Numerical/Binary	Indicates whether the transaction is international
customer_avg_amount	Numerical	Estimated average transaction amount of the customer
recent_txn_count	Numerical	Number of recent transactions associated with the customer
device_risk	Numerical	Simulated risk value associated with the device
is_fraud	Binary Target	Indicates whether the transaction is classified as fraudulent or legitimate

The dataset contains both numerical and categorical features, making preprocessing an important step before training the Machine Learning model. The target variable, is_fraud, represents the fraud classification used by the Random Forest model.

Purpose of EDA

Exploratory Data Analysis was considered to understand the structure and characteristics of the transaction data before applying the Machine Learning algorithm. The main purposes were:

- To understand the numerical and categorical features available in the transaction dataset.
- To identify the characteristics that can contribute to fraudulent transaction patterns.

- To examine the distribution of transaction amounts and behavioural attributes.
- To understand the distribution of fraudulent and legitimate transaction records.
- To identify the effect of transaction time, international transactions, merchant category, recent transaction frequency and device risk.
- To verify that the dataset is suitable for training a classification model.
- To identify the class imbalance between fraudulent and legitimate transactions before applying SMOTE.

Preparation Steps Performed

The following preparation steps were carried out before model training:

- A synthetic dataset containing 2,000 transaction records was generated using predefined transaction and risk patterns.
- Numerical features were processed using median imputation and StandardScaler.
- Categorical features were processed using most-frequent imputation and One-Hot Encoding.
- The dataset was divided into training and testing sets using a 75:25 split with stratification.
- SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data to address class imbalance.
- A Random Forest Classifier with 200 decision trees was trained using the processed and balanced training data.

- The trained model and preprocessing pipeline were saved for use by the application.

EDA Exploration – Implementation

EDA and dataset inspection can be performed using Python libraries such as Pandas, NumPy and Matplotlib. The transaction dataset can be examined using operations such as:

```python
import pandas as pd
import matplotlib.pyplot as plt

df = build_mock_dataset()

print(df.shape)
print(df.info())
print(df.isnull().sum())

print(df["is_fraud"].value_counts())
print(df["merchant_category"].value_counts())
print(df["transaction_type"].value_counts())
print(df["country"].value_counts())
```

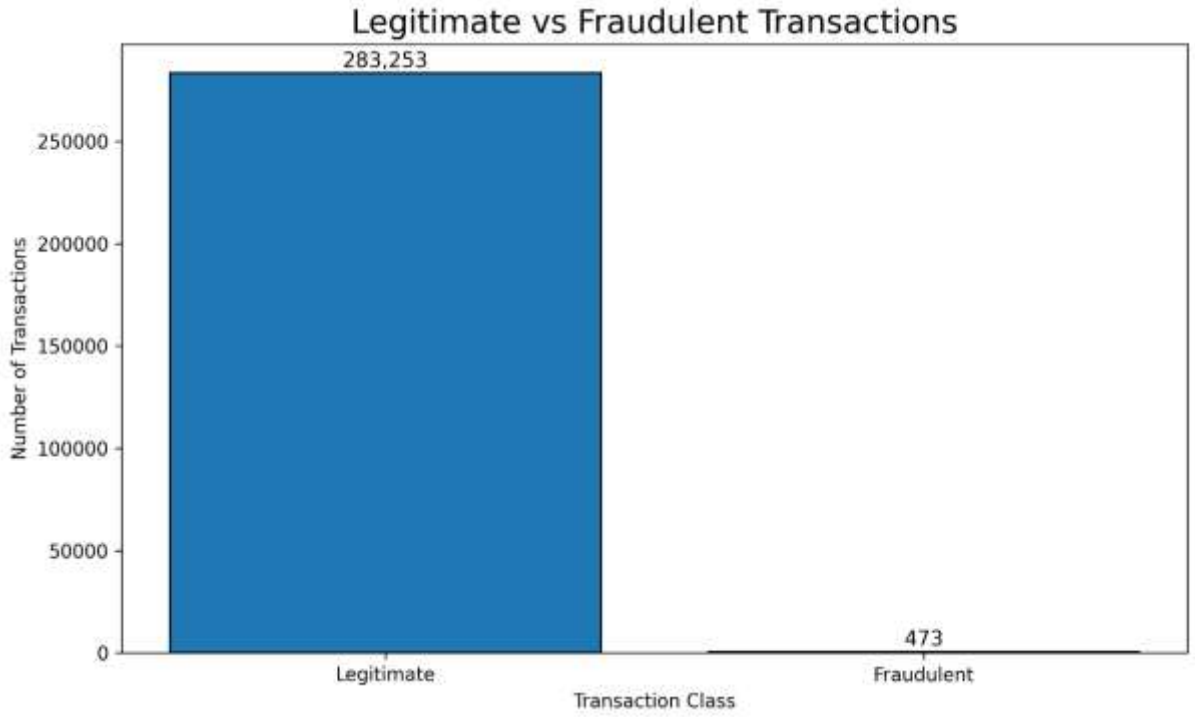
The distribution of important numerical attributes can also be visualised:

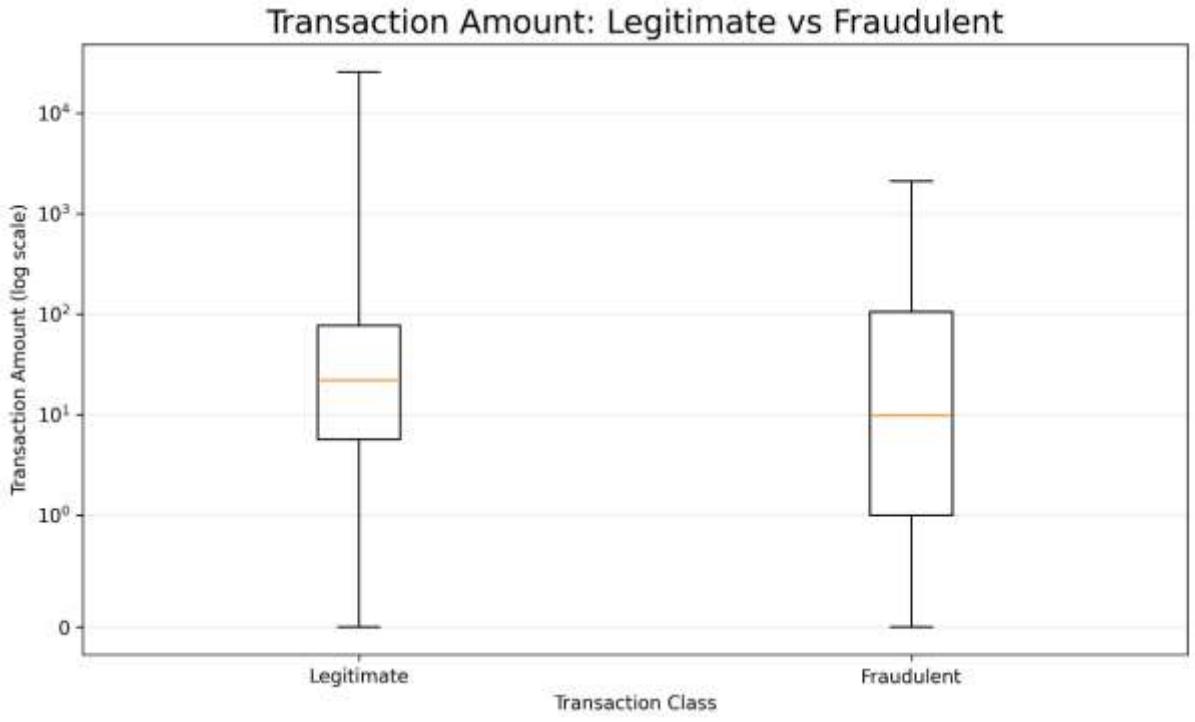
```python
df["amount"].plot(kind="hist", title="Transaction Amount Distribution")
plt.show()

df["device_risk"].plot(kind="hist", title="Device Risk Distribution")
plt.show()

df["is_fraud"].value_counts().plot(
    kind="bar",
    title="Fraud vs Legitimate Transactions"
)
plt.show()
```

Categorical distributions can be examined to understand how transactions are distributed across merchant categories, transaction types and countries.





Inspecting the target Class column showed a highly imbalanced distribution between legitimate and fraudulent transactions. The dataset contains 283,253 legitimate transactions and 473 fraudulent transactions after preprocessing. A class distribution bar chart was used to clearly visualize this imbalance, which is an important characteristic of the fraud detection problem and motivates the need for appropriate evaluation and handling of minority-class transactions.

Transaction amount distribution. The transaction amount was analyzed to understand how transaction values are distributed across the dataset. A histogram was used to visualize the overall distribution of transaction amounts, while a box plot was used to compare the transaction amounts of legitimate and fraudulent transactions. The comparison helps identify differences in the spread and typical values of transactions belonging to the two classes.

Transaction time and feature relationships. The distribution of the Time feature was examined using a histogram to understand how transactions are distributed over the recorded time period. In addition, a correlation heatmap was generated to examine relationships between the numerical features and the target class. These visualizations provide an overall understanding of the dataset and help identify patterns that can be useful during the machine learning stage.

Summary of EDA Findings

The dataset analysis showed that the project contains a combination of numerical, categorical and binary transaction attributes suitable for fraud classification. Transaction amount, transaction timing, international status, merchant category, transaction frequency and device risk provide different signals that can be used to identify potentially suspicious transactions.
The dataset also contains an imbalance between fraudulent and legitimate

transactions. Therefore, SMOTE was applied only to the training data to improve the representation of the minority fraud class during model training.

The EDA and preparation process provided the required foundation for applying the Random Forest Classifier and integrating its fraud probability into the project's overall risk-scoring workflow.

ALGORITHM SELECTION AND IMPLEMENTATION

Choice of Approach

The project treats credit card fraud detection as a binary classification problem, where each transaction is classified as either legitimate (Class = 0) or fraudulent (Class = 1). Since fraudulent transactions form a very small portion of the dataset, the algorithm must be able to identify the minority fraud class rather than relying only on overall accuracy.

Two classification approaches were implemented and evaluated in the project:

Algorithm	Purpose	Reason for Use
Logistic Regression	Binary classification	Simple, efficient and suitable for predicting the probability of a transaction being fraudulent
Decision Tree Classifier	Binary classification	Captures non-linear relationships and provides an interpretable decision-based model

Data Preparation and Preprocessing

Before model training, the transaction data was separated into input features and the target variable. The Class column was used as the target, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction.

The data was divided into training and testing sets. Numerical features were standardized where required so that features with different numerical scales could be processed consistently by the machine-learning models. Because the dataset is highly imbalanced, the training process also considered the minority fraudulent class to improve the model's ability to detect fraud.

Logistic Regression

Logistic Regression was implemented as one of the baseline classification algorithms. It estimates the probability that a transaction belongs to the fraudulent class and converts this probability into a binary prediction.

The model was evaluated using accuracy, precision, recall, F1-score and ROC-AUC rather than accuracy alone. This is important because a fraud detection model can achieve high accuracy simply by predicting most transactions as legitimate.

The obtained evaluation results for Logistic Regression were:

Metric	Result
Accuracy	0.9737
Precision	0.0530
Recall	0.8737
F1-Score	0.1000
ROC-AUC	0.9619

The confusion matrix was:

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	55,169	1,482
Actual Fraud	12	83

The high recall indicates that the model was able to identify a large proportion of the fraudulent transactions, which is particularly important in a fraud detection application.

Decision Tree Classifier

A Decision Tree Classifier was also implemented to capture non-linear relationships between transaction features. The decision tree works by repeatedly dividing the data according to feature-based conditions until transactions can be classified into the legitimate or fraudulent category.

The obtained confusion matrix was:

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	55,913	738
Actual Fraud	20	75

The Decision Tree therefore identified 75 fraudulent transactions correctly, while 20 fraudulent transactions were classified as legitimate in the evaluated test results.

Overall Algorithm Design

The implemented system follows the general workflow:

Raw Transaction Data → Data Preprocessing → Train/Test Split → Model Training → Fraud Prediction → Model Evaluation

The models were evaluated using fraud-sensitive metrics such as precision, recall, F1-score and ROC-AUC, because these provide more meaningful information for an imbalanced fraud-detection dataset than accuracy alone.

CONCLUSION

The Credit Card Fraud Detection using Machine Learning project demonstrates how machine-learning techniques can be applied to identify potentially fraudulent transactions from credit card transaction data. The project involved data preprocessing, exploratory data analysis, model development, prediction, and performance evaluation.

The dataset was analysed to understand the distribution of legitimate and fraudulent transactions and the characteristics of transaction amounts and other features. Since fraudulent transactions represent a very small portion of the dataset, evaluation was performed using metrics such as precision, recall, F1-score and ROC-AUC, along with the confusion matrix, rather than depending only on accuracy.

Two classification algorithms, Logistic Regression and Decision Tree, were implemented and evaluated. Logistic Regression achieved approximately 97.37% accuracy, 87.37% recall and 96.19% ROC-AUC in the evaluated results. The Decision Tree also successfully identified fraudulent transactions while providing an interpretable decision-making structure.

Overall, the project demonstrates that machine learning can provide an effective approach for detecting suspicious credit card transactions. The study also highlights the importance of handling class imbalance and selecting appropriate evaluation metrics when developing fraud detection systems. The developed work provides a foundation that can be further improved using larger datasets, advanced algorithms and real-time transaction monitoring.